Chapter 3

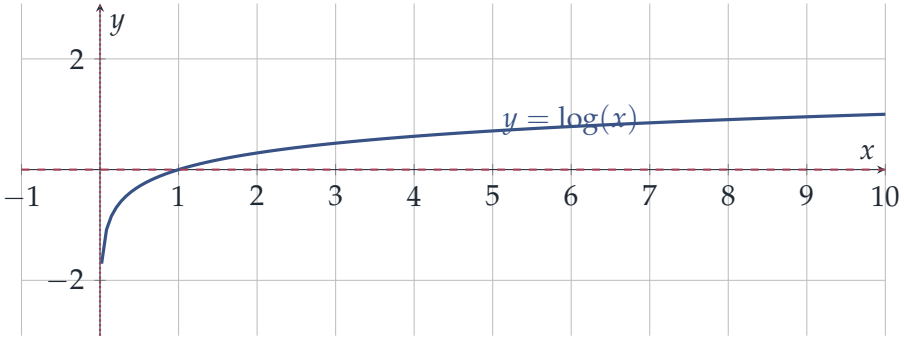
Decimal logarithm function

Competences: go from exponential writing to logarithmic writing, solve, model

3.1 Reference course

Key ideas

For every $b > 0$, $\log(b)$ is the unique real number x such as $10^x = b$. Thus $\log(10^x) = x$ and, for $b > 0$, $10^{\log(b)} = b$. The log function is defined on $]0, +\infty[$ and it is strictly increasing.



For $a > 0$, $b > 0$ and all natural n :

$$\log(ab) = \log(a) + \log(b), \qquad \log\left(\frac{a}{b}\right) = \log(a) - \log(b), \qquad \log(a^n) = n\log(a).$$

Frequent error

$\log(a+b)$ does not simplify. Before any logarithmic calculation, check that each argument is Strictly positive.

Method

To solve $a^x = b$, $a > 0$, $a \neq 1$ and $b > 0$, we take the logarithm: $x\log(a) = \log(b)$, therefore $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Solve 1 ,$08^x = 2$ is like writing $x = \dfrac{\log(2)}{\log(1{,}08)} \approx 9{,}01$. A value multiplied each year by So 1.08 doubles after just over nine years.

Like log is increasing, for $a, b > 0$, $a < b$ is equivalent to $\log(a) < \log(b)$. For $a > 1$, $a^x < b$ equals to $x < \dfrac{\log b}{\log a}$; when 0 $< a < 1$, the sense of inequality is inverse.

Session 15 — Definition of decimal logarithm

Decimal logarithm function

17/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Connect power of 10 and logarithm.

· Calculate normal values.

· Use the calculator.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 80 .....
2. Provide the log domain .....
3. Calculate log (10^4). .....
4. Resolve $10^x = 0{,}01$. .....
5. Developing log $(3b)$, $b > 0$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: connect power of 10 and logarithm and calculate Standard values.

Research activity

Completing couples $10^x = b$ and $x = \log(b)$ for $b = 0{,}001; 0{,}1; 1; 10; 1{,}000$. Then use the calculator for $b = 7$.

Trace written

For $b > 0$, $\log(b)$ is the only solution of $10^x = b$. In particular $\log(1) = 0$, $\log(10) = 1$ and $\log(100) = 2$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate log (10^2) and $\log(10^{-2})$.
2. Developing log $\left(\frac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve $2^x = 22$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, courses p. 64·69; exercises p. 70, n° 1 to 5.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 70, exercise 2

Solve the following equations.

a. $5 \times 10^{x} = 3{,}375$

b. $3{,}2 + 2 \times 10^{x} = 4{,}5 \times 10^{x}$

c. $-17{,}3 + 10^{x} = 5 - 3 \times 10^{x}$

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 16 — Meaning of Variation and Representation

Decimal logarithm function

19/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Know the field.

· Use growth.

· Read the curve.

Fluency · 6 to 8 minutes, without calculator

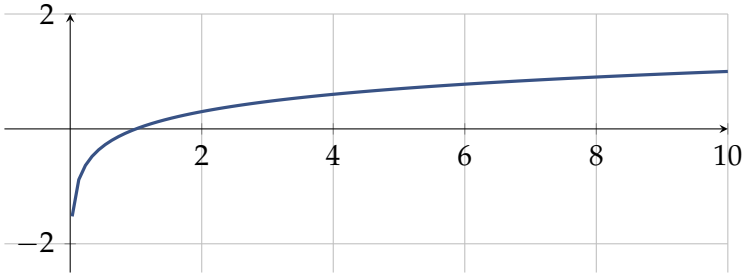
1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. Calculate log (10^0). . . . . .
3. Resolve 10 $^x = 0{,}01$. . . . . .
4. Developing log $(3b)$, $b > 0$. . . . . .
5. Provide the log domain . . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: know the domain and use growth.

Research activity

Trace $y = 10^x$ and the right $y = x$, then obtain the logarithm curve by symmetry. its domain, its sense of variation and asymptote.



Trace written

The function log is defined on $]0, +\infty[$, strictly increasing, and its curve admits $x = 0$ as vertical asymptote.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate $\log\left(10^3\right)$ and $\log(10^{-3})$.
2. Developing $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve $3^{x} = 23$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, courses p. 64·69; exercises p. 70, n° 6 to 10.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 70, exercise 7

Give the sign of the following numbers.

a. $\log(2{,}5)$ b. $\log(0{,}25)$ c. $\log\left(\dfrac{7}{10}\right)$

Personal work

Resuming Exercise 3 with the following data:
200 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 17 — Algebraic properties

Decimal logarithm function

24/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Develop or condense logarithms.

· Check conditions.

· Avoid false properties.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 6 - x + 2$. .....
2. Resolve 10 $^{x} = 0{,}01$. .....
3. Developing log $(3b), b > 0$. .....
4. Provide the log domain .....
5. Calculate log (10^{1}). .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before activity, reformulate keywords: develop or condense logarithms and check conditions.

Research activity

From 10 $^{2} \times 10^{3} = 10^{5}$, conjecture $\log(ab) = \log a + \log b$. Test numerically with $a = 2, b = 5$, then explain why a, b must be positive.

Trace written

For $a, b > 0$, $\log(ab) = \log a + \log b$ and $\log\left(\frac{a}{b}\right) = \log a - \log b$. More log $(a^{n}) = n \log a$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate log (10^{4}) and $\log(10^{-4})$.
2. Developing log $\left(\frac{100a^{2}}{b}\right)$ for $a, b > 0$.
3. Resolve 4 $^{x} = 24$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, courses p. 66·69; exercises p. 72, no. 20-23.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 73, exercise 24

Write in log form (A) where: A is a real number that will be specified, the following numbers:

a. $\log(2) + \log(7) - \log(5)$

b. $\log(3) - 2\log(5)$

c. $\log(3) + \log(7)$

d. $3\log(7) - 7\log(3)$

e. $\log(12) - \log(4) + 2\log(3)$

Personal work

Resuming Exercise 2 with the following data:
205 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 18 **Solve** $10^x = b$

Decimal logarithm function 26/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Resolve $10^x = b$.

· Framework a solution.

· Interpret an order of magnitude.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 140 .....
2. Developing $\log(3b)$, $b > 0$. .....
3. Provide the log domain .....
4. Calculate $\log(10^2)$. .....
5. Resolve $10^x = 0{,}01$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before activity, reformulate keywords: solve $10^x = b$ and frame a solution.

Research activity

A device measures intensities $b = 0{,}04; 3{,}2; 850$. For each value, resolve $10^x = b$, give x one hundredth and check by calculating 10^x.

Trace written

Equation $10^x = b$ a solution only if $b > 0$; this solution is $x = \log(b)$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate $\log(10^5)$ and $\log(10^{-5})$.
2. Developing $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve $5^x = 25$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, p. 73, No. 30 to 32. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 73, exercise 32

Harpagon invested a sum of 24,000 €, with compound interest at an annual rate of 3,5% Some time after, he decides to close this account while the balance is 40 208 ,37 €. The purpose of the exercise is to determine the number of years of existence of the account.

1. Complete the incomplete algorithm so that it responds to the problem.
2. Determine the number of years of existence of the account.
3. Translate the algorithm into Python language and implant it into a Python editor. Compare the the result obtained with that of question 2.

Personal work

Resuming Exercise 3 with the following data:
210 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 19

Solve $a^x = b$

Decimal logarithm function

01/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Solve $a^x = b$.

· Use a logarithm quotient.

· Arrondir with sense.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase . . . . . .
2. Provide the log domain . . . . . .
3. Calculate $\log(10^3)$. . . . . .
4. Resolve $10^x = 0{,}01$. . . . . .
5. Developing $\log(3b)$, $b > 0$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: solve $a^x = b$ use a logarithm quotient.

Research activity

The number of subscribers is multiplied by 1,15 each month. Determine after how many months he'll have tripled. Put the equation, solve it with logarithms and interpret rounding.

Trace written

For $a > 0, a \neq 1$ and $b > 0$, $a^x = b$ is equivalent to $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate $\log(10^6)$ and $\log(10^{-6})$.
2. Developing $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve $6^x = 26$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, p. 74, Nos. 33 to 39. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 73, exercise 30

1. A first capital of 6,000 € is invested at a compound interest rate of 9%.
   Will this capital have doubled over the years?

2. A capital of 9,000 € is placed on the same day at the rate of 6%.
   Has the value of the first capital exceeded the second?

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 20 — Equations $x^a = b$ and simple inequalities

Decimal logarithm function

03/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve an unknown power.

· Treating inequality.

· Take into account the field.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 9 - x + 2$. .....
2. Calculate log (10^4). .....
3. Resolve 10 $^x = 0{,}01$. .....
4. Developing log $(3b), b > 0$. .....
5. Provide the log domain .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before activity, reformulate the keywords: solve an unknown power and treat a inequality.

Research activity

Solve on $[0, +\infty[: x^3 = 40$, then $x^4 = 16$. Then resolve 1 $,08^x < 1{,}5$ and explain the role of growth.

Trace written

To solve $x^n = b$take into account the field and gender n. For exponential inequality, the direction depends on the growth or decay of the function.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate log (10^2) and $\log(10^{-2})$.
2. Developing log $\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve 2 $^x = 22$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, pp. 75, No. 40 to 42. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 74, exercise 36

A capital of 15 500 € is invested with compound interest for four and a half years.
16 480 € is also invested in compound interest for 5 years and 3 months at an annual rate of 6%.
What the investment rate of the first capital must be for the capital at the end of the investment are identical?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Session 21 — Contextualized problems

Decimal logarithm function

08/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Model a situation.

· Choose an equation.

· Conclude in context.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 60 .....
2. Resolve $10^{x} = 0{,}01$. .....
3. Developing $\log(3b)$, $b > 0$. .....
4. Provide the log domain .....
5. Calculate $\log(10^{3})$. .....

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: model a situation and choose an equation.

Research activity

The sound level is $L = 10\log\left(\frac{I}{I_0}\right)$. Calculate L where $I = 10^{6} I_0$. Then determine the report $\frac{I}{I_0}$ for $L = 85$ dB.

Trace written

In a problem, define the variable, set the equation, solve, control the domain and conclude with unity. A digital solution without interpretation is incomplete.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate $\log(10^{3})$ and $\log(10^{-3})$.
2. Developing $\log\left(\frac{100a^{2}}{b}\right)$ for $a, b > 0$.
3. Resolve $3^{x} = 23$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, p. 74, No. 38·39, then pp. 75, No. 40 to 42. The exercise reproduced in this sitting is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 75, exercise 41

The dynamic of an image is the ability of a digital image to return the scales light intensities of the real world. Dynamics are often expressed using stops.
In digital imaging, the darkest pixel (black) corresponds to 0 and the brightest pixel (white) There are 256 luminous intensities, which translates into computer units.
by 8 bits of precision (256) $= 2^8$The number of stops is: $\frac{\log(256)}{\log(2)} = 8.$

1. For an underexposed digital photo, only 128 luminous intensities can be How many stops does that match?
2. A RAW file has an accuracy of 10 bits. What is the number of corresponding stops?
3. A standard video camera has a dynamic image of 5 ,5 stops. How many intensity Does it capture different light?
4. The latest generation television screens have a dynamic image of 16 stops. How many different luminous intensities can they display?
5. The number of luminous intensities in the real world is around 100 000 000. Stop it, does it match?

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Session 22 — Assessment and evaluation

Decimal logarithm function

10/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Synthesize methods.

· Identify conditions.

· Remedy errors.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 25 per cent increase . . . . . .
2. Developing $\log(3b), b > 0$. . . . . .
3. Provide the log domain . . . . . .
4. Calculate $\log(10^4)$. . . . . .
5. Resolve $10^x = 0{,}01$. . . . . .

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: synthesize the methods and identify the conditions.

Research activity

Make a method map: domain; properties; equations $10^x = b$; equations $a^x = b$; Inequalities. Associate a condition and an example to each branch.

Trace written

Before a logarithm calculation: check the positivity. Before a comparison: identify the meaning of Before a conclusion: control likelihood.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Training · essential

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Developing $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Resolve $4^x = 24$ and give an approximate value to the hundredth.

Reference in the manual

Supplements: Calao, pp. 76-79. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources

Numbers of the manual.

Manuel Calao · p. 75, exercise 42

N is a natural non-zero integer. The purpose of the exercise is to find a method of determine the number of digits in its decimal entry.

1. $N = 10\,203$.

   a. Framework N between two powers of 10 consecutive.
   $\log(N)$ between two consecutive integers.

   b. Deduct the value rounded by excess to the unit near $\log(N)$. Compare this result with the number of digits of N.

2. N has 23 digits.

   a. Framework N between two powers of 10 consecutive.
   $\log(N)$ between two consecutive integers.

   b. Give the value rounded by excess to the unit near $\log(N)$. What value is there?
   ?

3. Deduce from previous questions a method for determining the number of each non-zero natural integer when given in decimal form.

4. Use the previous method to determine the number of digits of the following integer: a. 7^{49} ; b. 5658 ; c. $2\,019 \times 10^{20}$.

5. The largest number known to date is a number of Mersennes which writes: $2^{82\,589\,933} - 1$. How many numbers does he have?

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.